

Sewon Kim

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Seoul, South Korea

OBJECTIVE

Optimal Control and Robot Learning; Mathematically Sophisticated Theory into the Pysical World.

EDUCATION

- **University of Seoul** Mar 01 2022 – Present
B.E. in Electrical and Computer Engineering, Minor in Mathematics Seoul, South Korea

EXPERIENCE

- **Institute of Innovation for Future Army** Jan 13 2025 – Jul 12 2026 (Expected)
Drone Researcher (Compulsory Military Service) Daejeon, South Korea
- **InBody** Jan 05 2024 – Jan 12 2025
AI Lead Engineer, InBody AI Scale Seoul, South Korea
- **Control and Dynamic System Lab** Jan 31 2023 – Dec 23 2024
Fifteenth Lead Researcher; under the supervision of Prof. Gyunghoon Park Seoul, South Korea

PROJECTS

- **Autonomous air purifying robot agent** Aug 25 2025 – Sep 25 2026
4th Place (200 teams) — SK NAMUH X A1: Autonomous Driving Path Optimization (\$2,200) Seoul, South Korea
independently implemented full-stack navigation purely in NumPy, including MCL, MPPI, TSP.
- **Autonomous racing "F1tenth"** Mar 3 2024 – Dec 19 2024
5th Place (12 teams) — 22nd F1tenth Autonomous Grand Prix at CDC 2024 Milan, Italy
led project with two undergraduates; implemented SLAM, MCL, Raceline Opt, MPPI.
- **Cart pole system design and control by PID and NMPC** Jul 25 2024 – Sep 27 2024
3rd Place (25 teams) — UOS ECE Innovation Fair (\$220) Seoul, South Korea
independently developed 3D modeling, motor control, sensor integration, state estimation, MPC.
- **New fully-automated body composition analyzer "InBody AI Scale"** Jan 1 2024 – Jan 11 2025
Consumer Electronic Show 2025 Las Vegas, USA
developed height estimation (99.4%) and face recognition.
- **ECG-based cardiovascular diagnosis device using ML** Jul 1 2023 – Jul 31 2023
3rd Place (12 teams) — InBody Hackathon 2023 (\$1450) Seoul, South Korea
- **Autonomous mobile robot for executing missions** Jul 24 2023 – Jul 28 2023
3rd Place (20 teams) — AI Object Detection-Based Solar Tracking Robot Challenge (\$220) Seoul, South Korea
- **Dating app "Sidaeting"** Mar 7 2022 – Jun 31 2022
1,700 User Base Seoul, South Korea
developed the frontend for the dating service.

SKILLS

- PyTorch, C, C++, Java, Python, Javascript, React.js, Html, Css, Sql
- ROS1 Noetic, ROS2 Foxy · Humble, Mujoco, Matlab, Docker

LANGUAGES

- Korean
- English